DAC-LOOP: DISTILLATION-AWARE COMPRESSION LOOP WITH TEACHER-STUDENT LEARNING FOR REMOTE SENSING CNN

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ABSTRACT

Convolutional Neural Networks achieve state-of-the-art performance in remote sensing scene classification but remain challenging to deploy on resource-constrained devices. To address this challenge, we investigate how far a knowledge-distilled student model can be further compressed while preserving teacher-level accuracy. We propose a **Distillation-Aware Compression Loop (DAC-Loop)** that combines knowledge distillation and iterative pruning, allowing the distilled student to be compressed further and realigned with the teacher at each step. Experiments on EuroSAT and NWPU-RESISC45 show that the DAC-Loop consistently shifts the accuracy–compression trade-off in favor of smaller yet high-performing models: it delivers up to 3 percentage-point accuracy gains while reducing the parameter count by more than 90%, maintaining competitive accuracy even at high compression levels. These results demonstrate that combining distillation and pruning in a unified iterative loop offers an effective path toward compact and deployable remote sensing models.

***Index Terms*—** remote sensing, model compression, knowledge distillation, pruning, teacher-student learning

1. INTRODUCTION

Remote Sensing (RS) has benefited greatly from advances in deep neural networks, enabling highly accurate land-use and land-cover (LULC) classification on large-scale datasets such as EuroSAT and NWPU-RESISC45 [1]. LULC-classification became a central task for monitoring, environmental analysis, and constrained devices decision-making. However, deploying such models on resource-constrained devices remains challenging as state-of-the-art CNN-backbones require substantial memory, compute, and power [2].

This challenge motivates the development of compact yet accurate RS classifiers. While lightweight architectures exist, they rely on custom, task-specific designs and extensive architecture engineering [3]. A promising direction to mitigate this gap lies in teacher-student knowledge distillation (KD), where a high-capacity teacher model guides a smaller student model in training to achieve comparable accuracy [4] [5] [6]. Yet, even smaller models such as distilled students can exceed the tight memory and compute limits of resource-constrained platforms [7]. The goal of this paper is to explore how much a knowledge-distilled student model can be further compressed while preserving teacher-level accuracy on RS scene classification. In particular, our main contributions are:

- We introduce a Distillation-Aware Compression Loop (Figure 1) tailored to CNN backbones and evaluated on EuroSAT and NWPU-RESISC45.
- Ablation studies demonstrate that iterative DAC-Loop cycles outperform standard pruning on every level of compression (Figure 2).

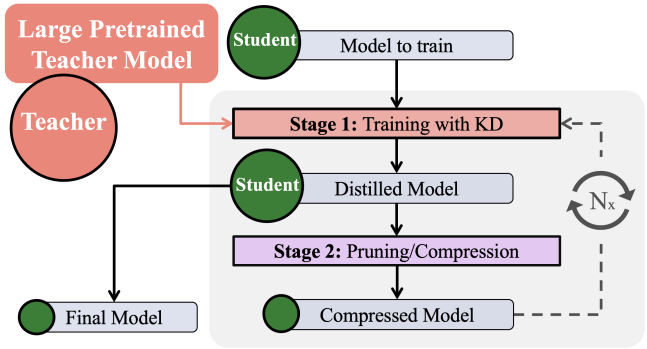


Fig. 1. **Distillation-Aware Compression (DAC) Loop**: Alternates between *Stage 1: KD-based training* of the student and a *Stage 2: pruning step* that compresses the distilled student further. After each pruning stage, the student is realigned with the teacher through another KD pass. A compression scheduler controls the pruning type and sparsity per iteration (N), enabling progressive multi-cycle compression.

2. RELATED WORK

Recent research explored lightweight architectures [3] and distillation techniques [4] for RS data, showing that distilled students can approach teacher-level accuracy while significantly reducing model size. Previous works further demonstrate clear accuracy–efficiency trade-offs, where reductions in parameter count or FLOPs through pruning or architectural simplification often incur proportional drops in classi-

fication performance. For example, Shinde et al. [8] illustrate this trade-off by pruning and quantizing standard CNN backbones on EuroSAT, showing that even substantial parameter and FLOP reductions result in only modest accuracy losses. Despite these advances, existing approaches typically treat distillation and pruning as separate steps. Prior work either (i) applies KD to train compact students without any subsequent pruning [4, 5, 6, 2] or (ii) compresses supervised RS models through pruning without leveraging distillation to recover lost performance [8, 9]. In contrast, our work integrates KD directly into the pruning process in order to retain teacher-level accuracy even at high sparsity levels. Moreover, we introduce a model agnostic framework to create a scheduled model compression through multiple dynamic loop iterations.

3. METHODOLOGY

3.1. Knowledge Distillation

In the first KD stage the predictive knowledge of a teacher model is transferred to a smaller student. Following Hinton et al. [10], the student is trained using a hybrid distillation objective. Let T_ϕ and S_θ denote the teacher and student networks, respectively. The KD objective is defined as

$$L = (1-\alpha)\, L_{\mathrm{CE}}(y, S_\theta(x)) + \alpha\, L_{\mathrm{KD}}(T_\phi(x), S_\theta(x)), \quad (1)$$

where α controls the trade-off between supervised learning and teacher guidance. We evaluate four KD formulations that differ in the definition of L_{KD}, as detailed in the appendix. For the *KD via Cosine Embedding Loss* variant, the *teacher* and *student* feature representations must have matching dimensionality, since the cosine loss is defined on pairs of vectors in the same space. Let $\mathbf{f}^{(t)} \in \mathbb{R}^{d_t}$ and $\mathbf{f}^{(s)} \in \mathbb{R}^{d_s}$ denote the intermediate feature vectors of the teacher and student, respectively. To enable a well-defined similarity computation when $d_t \neq d_s$, a learnable projection layer was introduced which maps the student features into the teacher's feature space. The resulting *cosine-embedding KD loss* is then computed between $\mathbf{f}^{(t)}$ and $\mathbf{f}^{(s)}_{\mathrm{proj}}$, ensuring shape compatibility while still allowing architectural flexibility between teacher and student models.

3.2. Pruning

To reduce the parameter count of the network, two complementary pruning strategies were implemented: (1) global unstructured pruning at weight level and (2) structured pruning at channel level for convolutional layers.

3.2.1. Global Unstructured Pruning

Let $\{\mathbf{W}_\ell\}_{\ell=1}^{L}$ denote the set of learnable weight tensors of all prunable layers (convolutional and linear) in the network. We introduce binary masks $\{\mathbf{M}_\ell\}_{\ell=1}^{L}$, with the same shape as $\mathbf{W}_\ell$, and define the effective weights as

$$\widetilde{\mathbf{W}}_\ell = \mathbf{M}_\ell \odot \mathbf{W}_\ell, \quad (2)$$

where $\odot$ denotes elementwise multiplication and $(\mathbf{M}_\ell)_i \in \{0,1\}$ encodes whether the i-th parameter is kept (1) or pruned (0). Given a desired global sparsity level $\alpha \in [0,1)$, which indicates the pruning fraction α of all weights across all prunable layers, we collect all scalar weights into a single multiset. For magnitude-based pruning, we compute a global threshold τ such that a fraction α of elements in $\mathcal{W}$ satisfies $|w| \leq \tau$. The mask is then defined as

$$(\mathbf{M}_\ell)_i = \begin{cases} 0, & \text{if } \left|(\mathbf{W}_\ell)_i\right| \leq \tau, \\ 1, & \text{otherwise.} \end{cases} \quad (3)$$

This mask $(\mathbf{M}_\ell)_i$ is applied during training via a standard pruning re-parameterization, ensuring that pruned weights no longer contribute to the forward pass.

3.2.2. Structured Channel Pruning

While global unstructured pruning increases weight-level sparsity, it does not change the layer shapes and thus does not directly reduce the number of channels. To obtain a structurally smaller network, we additionally perform structured pruning of convolutional channels following the channel-pruning formulations introduced by Li et al. [11] and He et al. [12]. Consider a convolutional layer ℓ with weight tensor

$$\mathbf{W}_\ell \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k_h \times k_w}, \quad (4)$$

where each slice $\mathbf{W}_{\ell,i,:,:,:}$ corresponds to the kernel associated with output channel $i \in \{1, \ldots, C_{\text{out}}\}$. We define the importance score of channel i as the ℓ_2 norm of its kernel:

$$s_{\ell,i} = \|\mathbf{W}_{\ell,i,:,:,:}\|_2^2 = \sum_{c=1}^{C_{\text{in}}} \sum_{u=1}^{k_h} \sum_{v=1}^{k_w} \left(\mathbf{W}_{\ell,i,c,u,v}\right)^2. \quad (5)$$

For a defined pruning ratio $r_\ell \in [0,1)$ for layer ℓ, we compute $n_\ell = \lfloor r_\ell C_{\text{out}} \rfloor$ and prune the n_ℓ output channels with the smallest scores $s_{\ell,i}$. In practice, removing channels from one layer affects the input dimensionality of subsequent layers. To ensure architectural consistency, a dependency graph of the network is constructed. Pruning is then applied to the entire group atomically, guaranteeing that all tensor shapes remain compatible and the resulting network is a valid, smaller architecture.

3.3. Distillation–Aware Compression Loop (DAC)

As illustrated in Fig. 1, a *teacher* model $f_T(\cdot;\theta_T)$, in this work a ResNet50, is trained and frozen throughout the complete DAC-Loop. A smaller *student* network $f_S(\cdot;\theta_S)$, in this

work a ResNet18, is randomly initialized and trained with the defined KD-method and the pretrained *teacher* model in *Stage 1: Training with KD*. The distilled student model obtained from Stage 1 is then passed to *Stage 2: Pruning / Structural Compression*, where it is gradually reduced in size through structured pruning, altering the architecture by removing entire channels or filters. This requires a subsequent fine-tuning step to recover performance, which is again performed using the KD-based training procedure from Stage 1, where the compressed model continues to benefit from the teacher's guidance. After pruning and distillation-based recovery, the student re-enters the loop, allowing the process to run for *N* compression cycles. To support multiple DAC iterations, we employ a compression scheduler that assigns specific pruning strategies, hyperparameters and compression ratios to each loop cycle. This enables heterogeneous compression across iterations, e.g. unstructured pruning in the first cycle and structured pruning in the next.

4. EXPERIMENTS & RESULTS

Model	EuroSAT	RESISC45
SOTA-Baselines		
ResNet18 [13]	98.06	92.10
ResNet50-InDomain [14]	99.20	96.80
ResNet50 [15]	98.57	–
Our trained baselines		
TinyCNN	89.12± 0.12	28.60± 1.03
ResNet10mini	86.41± 0.09	83.50± 0.38
ResNet10	98.36± 0.05	88.07± 0.32
ResNet18	98.34± 0.05	90.29± 0.28
ResNet50	96.79± 0.18	92.72± 0.06

Table 1. Our baseline validation accuracy compared to SOTA-Baselines: baselines were trained for 50 epochs (except ResNet50 on NWPU-RESISC45 trained for 100 epochs).

4.1. Data & Setup

To demonstrate the DAC-Loop, *EuroSAT* [15] and *NWPU-RESISC45* [16] where used as datasets. The EuroSAT dataset consists of 27,000 Sentinel-2 images of size 64×64 distributed across ten LULC-classes. NWPU-RESISC45 comprises 31,500 RGB images of size 256×256, evenly distributed across 45 scene categories. To train the models, we adopt a 60-20-20 split. Standard normalization is applied across all experiments, while data augmentation is employed only for training the ResNet50 teacher model. All experiments were conducted on an NVIDIA DGX V100-SXM3-32GB. We trained all models using the AdamW optimizer with an initial learning rate of 3×10^{-4} and a batch size of 256, evaluating on the validation set after every epoch. Unless stated otherwise, the learning rate was scheduled using a cosine annealing schedule over the full 50 training epochs.

4.2. Baselines

Table 1 reports the model baselines we trained from scratch and compares them to published state-of-the-art baselines in prior work [13, 14, 15]. The results show that our implementations reach comparable accuracy on both EuroSAT and NWPU-RESISC45, providing a reliable and well-calibrated reference point for the compression experiments that follow.

4.3. Ablation Study

ResNet-18 Ablation Study Results on RESISC45						
	KD-1	KD-2	N	Comp.	Params	Val. Acc.
Base	–	–	1	0%	11.2 M	90.29± 0.28
	–	–	1	0%	2.0 M*	83.50± 0.38
	✓	–	1	0%	11.2 M	92.85± 0.15
Comp. +	–	–	1	47.5%	5.8 M	90.32± 0.14
	✓	–	1	47.5%	5.8 M	91.26± 0.07
	–	✓	1	47.5%	5.8 M	92.06± 0.11
	✓	✓	1	47.5%	5.8 M	92.68± 0.24
Comp. ++	–	–	1	75.8%	2.7 M	88.60± 0.15
	✓	–	1	75.8%	2.7 M	89.52± 0.33
	–	✓	1	75.8%	2.7 M	90.39± 0.12
	✓	✓	1	75.8%	2.7 M	91.23± 0.16
Loop	✓	✓	6	85.2%	1.6 M	**92.00**± 0.35
	✓	✓	9	92.1%	0.8 M	90.71± 0.25

* 2.0M model: ResNet10mini, our own small-model baseline.

Table 2. **KD-1 & KD-2** both represent **Stage 1 Training with KD** in our DAC-Loop (Figure 1) and indicate whether KD is applied pre (1) or post (2) structured channel pruning. **N** denotes the number of DAC-Loop iterations, **Comp.** the parameter compression ratio compared to the full baseline (0% compression).

In Table 2, the results of our DAC-Loop ablation study are grouped into four blocks: *Base* (no compression, our baselines), *Comp.+* (moderate structured compression strength at 38.43% parameter reduction), *Comp.++* (stronger structured compression strength at 65.71%), and *Loop* (multiple DAC iterations leading to even higher compression). Within each block, we systematically test the KD components, and vary the number of loop iterations N.

Across both structured compression degrees (*Comp.+* and *Comp.++*), the grey-highlighted rows show that the combination of KD-1 and KD-2 consistently yields the best performance. This confirms that integrating pre-pruning KD with post-pruning KD in structured channel pruning is more effective than either approach alone. The *Loop* experiments further illustrate the benefit of iterating the DAC-Loop: with

$N = 6$ iterations we obtain a model compressed to 1.6M parameters (85.23% compression) that still achieves the highest overall accuracy (92.44%, bold in Table 2). This ablation study demonstrates that repeated DAC-Loop cycles enable progressively higher compression while preserving, and even enhancing, model accuracy.

ResNet-18 unstructured pruning on NWPU-RESISC45					
KD-1	**KD-2**	**N**	**Comp.**	**Params**	**Val. Acc.**
✓	✓	1	0%	11.2 M	93.21± 0.00

Table 3. ResNet-18 student after applying global unstructured magnitude pruning on NWPU-RESISC45.

Table 3 shows that applying global unstructured magnitude pruning surprisingly boosts accuracy to 93.21%, outperforming all structured-pruning configurations in Table 2. Since unstructured pruning removes individual weights rather than whole channels, it does not change the model architecture or reduce parameter count. Although this sparsity does not translate into deployment efficiency, it provides accuracy benefits that structured pruning alone does not achieve. This motivates combining both complementary pruning methods within our DAC-Loop to benefit from compression and generalization advantages.

4.4. Results and Discussion

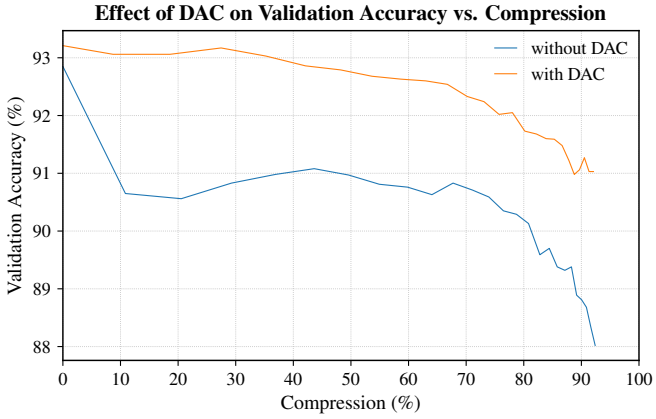


Fig. 2. **Effect of DAC-Loop on the accuracy–compression trade-off**: comparison of validation accuracy under increasing compression for ResNet-18 with (orange) and without (blue) the proposed DAC-Loop. Across all compression levels, the DAC-Loop maintains consistently higher accuracy than standard pruning.

Figure 2 illustrates the effect of the DAC-Loop on the accuracy–compression trade-off for ResNet-18 on NWPU-RESISC45. Across the entire compression range, the DAC curve remains consistently above the purely pruned baseline, demonstrating that distillation-guided recovery stabilizes accuracy. The difference can constantly be observed over all compression rates between 10-90%. This confirms that the DAC-Loop does not only mitigate pruning loss, it fundamentally shifts the trade-off, enabling significantly smaller models to maintain strong performance.

Table 4. Inference time and model size comparison between the full (11M) and compressed (0.8M) models on the RESISC45 validation split.

Metric	**Full**	**Compressed**	**Speedup**
Batch (ms) ↓	281.46	65.25	×4.3
Per-sample (ms) ↓	1.11	0.25	×4.3
Size (MB) ↓	128.19	3.58	×35.8

Additionally, Table 4 shows that the compressed model achieves a 4.3× inference speedup and a 35.8× reduction in model size. This experiment uses the nine-iteration DAC-Loop model introduced in Table 2. These improvements mark a critical advancement for on-device deployment, enabling remote-sensing classifiers to operate within the strict memory and latency constraints.

5. CONCLUSION

In this work we introduced a model-agnostic Distillation-Aware Compression Loop that combines knowledge distillation with structured and unstructured pruning for remote sensing CNNs. The loop alternates between KD-based training and pruning, with a compression scheduler controlling the pruning type and sparsity at each iteration, so that students can be progressively compressed and realigned with their teacher. On EuroSAT and NWPU-RESISC45, the DAC-Loop turns ResNet students into compact yet high-performing models. For a ResNet-18 student on NWPU-RESISC45, our method reduces parameter count by over 90% while matching or exceeding the supervised baseline. The ablation study shows that KD before and after structured channel pruning consistently outperforms pure pruning, and that iterative DAC cycles yield highly compressed models that still maintain strong accuracy. Global unstructured magnitude pruning further improves generalization without changing the architecture, and combining it with structured pruning inside the DAC-Loop allows us to harvest both benefits, regularization and true architectural compression.

The compression–accuracy curves confirm that the DAC-Loop systematically shifts the trade-off in favor of smaller models: across the entire compression range, DAC-compressed networks remain more accurate than their purely pruned counterparts. Overall, our results indicate that distillation-aware compression is a promising route toward deploying accurate, lightweight remote sensing classifiers on resource-constrained platforms.

6. REFERENCES

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7. APPENDIX

7.1. Knowledge Distillation Methods

In this section the detailed equations of the implemented KD-methods are explained.

7.1.1. KD via Cross-Entropy Loss

We employ classical soft-label distillation using temperature-scaled logits. Teacher and student predictions are softened using a temperature T_{temp}, and the distillation term is given by

$$L_{\text{KD}}^{\text{CE}} = \text{CE}\left(\text{softmax}\left(\frac{T_\phi(x)}{T_{\text{temp}}}\right), \text{softmax}\left(\frac{S_\theta(x)}{T_{\text{temp}}}\right)\right). \tag{6}$$

This encourages the student to match the teacher's softened class distribution.

7.1.2. KD via KL Divergence

Soft-label distillation can also be expressed using Kullback–Leibler divergence:

$$L_{\text{KD}}^{\text{KL}} = T_{\text{temp}}^2 \,\text{KL}\left(\text{softmax}\left(\frac{T_\phi(x)}{T_{\text{temp}}}\right) \,\middle\|\, \text{softmax}\left(\frac{S_\theta(x)}{T_{\text{temp}}}\right)\right). \tag{7}$$

The T_{temp}^2 scaling follows standard practice to preserve gradient magnitude.

7.1.3. KD via Cosine Embedding Loss

To guide the student towards the teacher's internal representations, we additionally evaluate feature-based distillation. Let $f_T(x)$ and $f_S(x)$ denote teacher and student embeddings from the penultimate layer. The distillation term becomes

$$L_{\text{KD}}^{\cos} = 1 - \cos(f_T(x),\, f_S(x)), \tag{8}$$

which penalizes angular deviations between embeddings.

7.1.4. KD via Logit Regression (MSE)

Finally, we consider logit-level regression, where the student directly regresses the teacher's logits:

$$L_{\text{KD}}^{\text{MSE}} = \|T_\phi(x) - S_\theta(x)\|_2^2. \tag{9}$$

This variant provides smooth gradients and is particularly effective for small-capacity students.

7.2. Ablation Study Results

KD-1 & KD-2 both represent **Stage 1 Training with KD** in our DAC-Loop (Figure 1) and indicate whether KD is applied pre (1) or post (2) pruning. **N** denotes the number of DAC-Loop iterations, **Comp.** the parameter compression ratio compared to the full baseline (0% compression).

ResNet-10 Ablation Study Results on RESISC45

	KD-1	KD-2	N	Comp.	Params	Val. Acc.
Base	–	–	1	0%	4.9 M	88.31± 0.00
	–	–	1	0%	2.0 M*	83.30± 0.00
	✓	–	1	0%	4.9 M	90.98± 0.00
Comp. +	–	–	1	38.4%	3.0 M	90.14± 0.00
	✓	–	1	38.4%	3.0 M	89.02± 0.00
	✓	✓	1	38.4%	3.0 M	91.16± 0.00
	✓	✓	1	38.4%	3.0 M	91.00± 0.00
Comp. ++	–	–	1	65.7%	1.6 M	89.52± 0.00
	✓	–	1	65.7%	1.6 M	89.62± 0.00
	–	✓	1	65.7%	1.6 M	89.02± 0.00
	✓	✓	1	65.7%	3.0 M	89.97± 0.00
Loop	✓	✓	6	38.4%	3.0 M	**91.33**± 0.00
	✓	✓	9	61.9%	1.8 M	90.89± 0.00

* 2.0M model: ResNet10small, our own small-model baseline.

ResNet-10 Ablation Study Results on EuroSAT

	KD-1	KD-2	N	Comp.	Params	Val. Acc.
Base	–	–	1	0%	4.9 M	98.40± 0.00
	✓	–	1	0%	4.9 M	98.57± 0.00
Comp. +	–	–	1	38.5%	3.0 M	98.04± 0.00
	✓	–	1	38.5%	3.0 M	98.20± 0.00
	–	✓	1	34.6%	3.2 M	98.19± 0.00
	✓	✓	1	34.6%	3.2 M	98.24± 0.00
Comp. ++	–	–	1	65.7%	1.7 M	97.83± 0.00
	✓	–	1	65.7%	1.7 M	97.89± 0.00
	–	✓	1	63.1%	1.8 M	98.24± 0.00
	✓	✓	1	63.1%	1.8 M	98.20± 0.00

ResNet-18 Ablation Study Results on EuroSAT

	KD-1	KD-2	N	Comp.	Params	Val. Acc.
Base	–	–	1	0%	11.2 M	98.27± 0.00
	✓	–	1	0%	11.2 M	98.74± 0.00
Comp. +	–	–	1	47.5%	5.9 M	98.06± 0.00
	✓	–	1	47.5%	5.9 M	98.35± 0.00
	–	✓	1	46.0%	6.0 M	98.00± 0.00
	✓	✓	1	46.0%	6.0 M	98.56± 0.00
Comp. ++	–	–	1	75.8%	2.7 M	98.07± 0.00
	✓	–	1	75.8%	2.7 M	97.80± 0.00
	–	✓	1	75.0%	2.8 M	97.94± 0.00
	✓	✓	1	75.0%	2.8 M	98.33± 0.00

7.3. Results of KD experiments

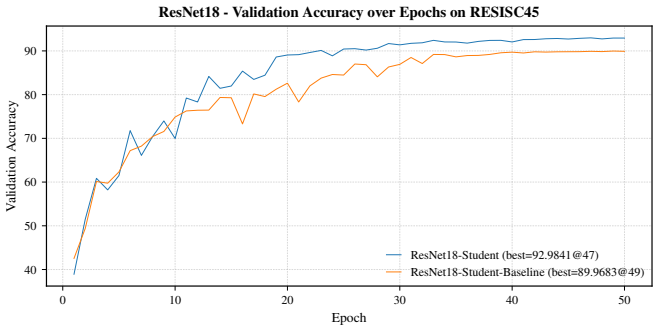


Fig. 3. ResNet18 Training with (blue) and without (orange) a Teacher